Earth, units and angles

$$1\ \text{nmi} = 1852\ \text{m} \approx 1'\ \text{of great-circle arc},$$

$$1^\circ = 60\ \text{nmi}, \quad 1\ \text{kt} = 1\ \text{nmi/h}$$

$$d_{\text{meridian}}\ [\text{nmi}] = \Delta\varphi\ [']\,, \qquad d_{\text{parallel}}\ [\text{nmi}] = \Delta\lambda\ ['] \cos\varphi$$

$$\delta = \theta_{\text{track}} - \theta_{\text{heading}}, \qquad \theta_{\text{true}} = \theta_{\text{mag}} + D$$

D > 0 if magnetic north is EAST of true north; runways and VOR radials are magnetic

Angles clockwise from north, three digits (090 = east, 270 = west). Heading: where the nose points. Track: path over the ground. Bearing: direction of an object from north; relative bearing: from the aircraft axis. Radial: direction away from the beacon.

Loxodrome (rhumb line): constant heading, straight on Mercator. Orthodrome (great circle): shortest, about straight on Lambert.

Wind triangle

$$\vec{V}_g = \vec{V}_a + \vec{V}_w, \qquad \theta_{w,\text{to}} = \theta_{w,\text{from}} + 180^\circ$$

air vector: heading and TAS; ground vector: track and GS

$$V_x = V\sin\theta\ (\text{east}), \quad V_y = V\cos\theta\ (\text{north}), \quad \theta = \operatorname{atan2}(V_x, V_y)$$

fix the quadrant from the signs of Vx and Vy

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}, \qquad c^2 = a^2 + b^2 - 2ab\cos C$$

$$\text{WCA} = \arcsin\frac{V_w\sin(\theta_{w,\text{to}} - \theta_{\text{trk}})}{\text{TAS}}, \qquad \theta_{\text{hdg}} = \theta_{\text{trk}} - \text{WCA}$$

heading to fly for a desired track: crab INTO the wind

$$\text{GS} = \text{TAS}\cos(\text{WCA}) + V_w\cos(\theta_{w,\text{to}} - \theta_{\text{trk}})$$

$$(V_I t)^2 = d^2 + (V_T t)^2 - 2\,d\,V_T t\cos\gamma, \qquad \sin\alpha = \frac{V_T}{V_I}\sin\gamma$$

γ: angle between the target track and the line of sight; α: lead angle

$$v = \frac{d}{\Delta t}, \qquad \Delta t \text{ in hours}$$

Signals

$$\langle x\rangle = \frac{1}{T_0}\int_{T_0} x\,dt, \quad P_m = \frac{1}{T_0}\int_{T_0} x^2\,dt,$$

$$x_{\text{rms}} = \sqrt{P_m}, \quad E = \int_{-\infty}^{\infty} x^2\,dt$$

$$A\cos(2\pi f_0 t):\ \langle x\rangle = 0,\ P_m = \frac{A^2}{2},\ x_{\text{rms}} = \frac{A}{\sqrt{2}}$$

$$A\cos(2\pi f_0 t + \varphi) = \frac{A}{2}e^{j(2\pi f_0 t+\varphi)} + \frac{A}{2}e^{-j(2\pi f_0 t+\varphi)}$$

two-sided spectrum: two lines of A/2 at +f0 and -f0

$$-A\cos x = A\cos(x + 180^\circ), \qquad A\sin x = A\cos(x - 90^\circ)$$

$$x(t) = \sum_n c_n e^{j2\pi n f_0 t}, \quad c_n = \frac{1}{T_0}\int_{T_0} x(t)e^{-j2\pi n f_0 t}dt,$$

$$c_0 = \langle x\rangle, \quad P_m = \sum_n |c_n|^2$$

$$\text{square wave } \pm 1:\ x(t) = \frac{4}{\pi}\sum_{k\geq 1}\frac{\sin\big((2k-1)2\pi f_0 t\big)}{2k-1}$$

$$\text{pulse train } (A,\tau,T):\ c_n = \frac{A\tau}{T}\operatorname{sinc}\frac{n\tau}{T}, \qquad \operatorname{sinc} x = \frac{\sin\pi x}{\pi x}$$

lines every 1/T, first zero of the envelope at f = 1/τ, mean = A τ/T

$$A\,\Pi(t/\tau) \longleftrightarrow A\tau\operatorname{sinc}(f\tau), \qquad E = \int |x|^2 dt = \int |X(f)|^2 df$$

$$X_k = \sum_{n=0}^{N-1} x_n e^{-j2\pi kn/N}$$

FFT = fast algorithm for the DFT, about N log2 N operations instead of N^2

Systems and modulation

$$Y(f) = X(f)H(f), \quad y(t) = x(t) * h(t) = \int x(\lambda)h(t-\lambda)\,d\lambda,$$

$$h = \mathcal{F}^{-1}\{H\}$$

$$H(f) = K\ (|f| < B) \ \Rightarrow\ h(t) = 2BK\operatorname{sinc}(2Bt)$$

$$y = Kx(t - t_0) \Leftrightarrow H = Ke^{-j2\pi f t_0},$$

$$t_p = -\frac{\arg H}{2\pi f}, \quad t_g = -\frac{1}{2\pi}\frac{d\arg H}{df}$$

|H| not flat → amplitude distortion; phase not linear in f → phase distortion (both linear, can be equalised). Bandwidth: range where |H| is flat, usually to -3 dB.

$$x(t)\cos(2\pi f_0 t) \longleftrightarrow \tfrac{1}{2}\big[X(f-f_0) + X(f+f_0)\big]$$

$$\cos^2\omega_0 t = \tfrac{1}{2} + \tfrac{1}{2}\cos 2\omega_0 t$$

non-linear systems create NEW frequencies: harmonics n*f0, intermodulation ±k f1 ± l f2

Needed bandwidth: voice 300-3400 Hz (3.1 kHz), video 5 MHz, digital r symbols/s needs at least r/2 Hz.

Noise, decibels, capacity

$$N = kTB, \quad k = 1.38\times 10^{-23}\ \text{J/K};$$

$$T = 290\ \text{K}:\ N\ [\text{dBm}] = -174 + 10\log_{10} B$$

$$G\ [\text{dB}] = 10\log\frac{P_2}{P_1}, \quad P\ [\text{dBm}] = 10\log\frac{P}{1\ \text{mW}},$$

$$\frac{S}{N}\ [\text{dB}] = S\ [\text{dBm}] - N\ [\text{dBm}]$$

x2 = 3 dB, x4 = 6 dB, x10 = 10 dB. 0 dBm = 1 mW, 30 dBm = 1 W, -30 dBm = 1 uW, -60 dBm = 1 nW, -90 dBm = 1 pW. Free space: doubling the distance costs 6 dB (S/N 20 dB → 14 dB).

$$C = B\log_2\left(1 + \frac{S}{N}\right)\ [\text{bit/s}]$$

e.g. 25 kHz, 30 dB: C = 249.2 kbit/s

Data transmission and A/D

$$r_{s,\max} = 2B\ [\text{baud}], \qquad r_b = r_s\log_2 M = n\,r_s\ [\text{bit/s}], \qquad M = 2^n$$

Same S/N and baud: more levels → more errors. Error probability: PSK < FSK < ASK. Binary synchronous r_b = r_s; asynchronous r_b < r_s (start/stop); M-ary r_b > r_s.

$$d_{\min} = d:\ \text{detects } d-1, \quad \text{corrects } \left\lfloor\tfrac{d-1}{2}\right\rfloor$$

parity: detects 1, corrects 0; Hamming (7,4): d = 3, corrects 1

Line codes: NRZ/RZ, unipolar/bipolar. AMI: no DC, but long zero runs lose sync; HDB3 replaces 4 zeros; Manchester: transition mid-bit, no DC, easy sync, more bandwidth.

A/D: anti-aliasing filter → sampler (Ts = 1/fs) → quantizer → coder. D/A: decoder + low-pass filter.

$$X_s(f) = f_s\sum_k X(f - kf_s), \qquad f_s \geq 2W$$

$$L = 2^N, \quad \Delta = \frac{V_{pp}}{2^N}, \quad T_b = \frac{T_s}{N}, \quad r_b = f_s N$$

$\text{voice: } 8\text{ kHz} \times 8\text{ bit} = 64\text{ kbit/s}, \qquad \text{SQNR} \approx 6.02N + 1.76\text{ dB}$

SQNR is the standard full-scale-sine result, not in the course notes

Waves and propagation

$$\lambda = \frac{c}{f}, \quad \eta = \sqrt{\frac{\mu}{\varepsilon}} = 377\ \Omega\ (\text{vacuum}),$$

$$v_p = \frac{1}{\sqrt{\mu\varepsilon}}, \quad k = \frac{\omega}{v_p}, \quad n = \sqrt{\varepsilon_r}$$

$$\theta_r = \theta_i, \qquad n_1 \sin\theta_i = n_2 \sin\theta_t$$

Polarization (tip of E): components in phase → linear; 90° apart → elliptical; 90° and equal amplitude → circular. Diffraction: wave bends round obstacles (Huygens).

Antennas and link budget

$$P_L = P_T D_T D_R \left(\frac{\lambda}{4\pi R}\right)^2 \prod_i c_i \qquad (\text{linear units}, 0 < c_i \le 1)$$

$$S = \frac{P_T D_T}{4\pi R^2}, \qquad P_L = S\, A_{ef}, \qquad A_{ef} = \frac{\lambda^2 D}{4\pi}$$

$$P_L\,[\text{dBm}] = P_T\,[\text{dBm}] + D_T + D_R - L_{fs} + \sum c_i\,[\text{dB}],$$

$$L_{fs} = 20 \log \frac{4\pi R}{\lambda}$$

$$R_{\max} = \frac{\lambda}{4\pi}\sqrt{\frac{P_T D_T D_R}{P_{L,\min}}}, \qquad R = \frac{\lambda}{4\pi} 10^{L_{fs}/20}$$

$$A_{ef} = 0.7\,\frac{\pi d^2}{4} \;\Rightarrow\; d = \sqrt{\frac{4A_{ef}}{0.7\pi}}$$

$$t(\theta,\varphi) = \frac{D(\theta,\varphi)}{D}, \quad G = e\,D,$$

$$\text{NLPS} = 0\text{ dB} - \text{largest side lobe}$$

beamwidth at -3 dB and between nulls; front-to-back: main lobe vs lobe at 180°

$$Z_{ant} = R_{rad} + R_\Omega + jX_{ant}, \qquad V_{ant} = l_{ef}E, \qquad A_{ef} = \frac{\eta\, l_{ef}^2}{4R_{ant}}$$

Isotropic: equal in all space (ideal). Omnidirectional: equal in one plane. Reciprocity: same pattern in Tx and Rx. Dipole λ/2, monopole λ/4: at 127.5 MHz λ = 2.35 m → 1.18 m / 59 cm.

Definitions asked again and again

Term	Meaning
Simplex / half-duplex / full-duplex	one way / both ways, one at a time (pilot-ATC) / both at once
Point to point / broadcast / radar	1 Tx to 1 Rx / few Tx, many Rx (NDB, GPS) / Tx and Rx together
Topologies	bus, ring, star, tree (bus of buses), hybrid
Telecommunication (ITU)	transmission of information by electromagnetic means
Modem	converts low-pass digital to band-pass digital and back
Crosstalk	one line's information leaks into another line of the same cable